सहायक प्राध्यापक (माहिती तंत्रज्ञान), महाराष्ट्र शिक्षण सेवा, गट-अ

(महाविद्यालयीन शाखा)

उच्च व तंत्र शिक्षण विभाग

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परीक्षेचे टप्पे: **१)** लेखी परीक्षा - २०० गुण **२)** मुलाखत- ५० गुण

-: परीक्षा योजना :-

विषय व सांकेतांक	माध्यम	प्रश्नसंख्या	गुण	कालावधी	दर्जा	प्रश्नपत्रिकेचे स्वरुप
विषयांशी संबंधीत घटक (११४७)	इंग्रजी	१००	२००	एक तास	पदव्युत्तर पदवी	वस्तुनिष्ठ बहुपर्यायी

अ) नकारात्मक गुणदान -

१) प्रत्येक चुकीच्या उत्तराकरीता २५% किंवा १/४ एवढे गुण एकूण गुणांमधून वजा/ कमी करण्यात येतील.
२) एखाद्या प्रश्नाची एकापेक्षा अधिक उत्तरे दिली असल्यास अथवा ज्या उमेदवाराने उत्तरपत्रिकेत पूर्ण वर्तुळ चिन्हांकित केले नसेल अशा प्रश्नाचे उत्तर चुकीचे समजण्यात येऊन त्या प्रश्नाच्या उत्तराकरीता २५% किंवा १/४ एवढे गुण एकूण गुणांमधून वजा/कमी करण्यात येतील.
३) वरीलप्रमाणे कार्यपध्दतीचा अवलंब करताना एकूण अंतिम गुणांची बेरीज अपूर्णांकात आली तरीही ती अपूर्णांकातच राहील व पुढील कार्यवाही त्याच्या आधारे करण्यात येईल.
४) एखाद्या प्रश्नाचे उत्तर अनुत्तरित असेल तर, अशा प्रकरणी नकारात्मक गुणांची पध्दत लागू असणार नाही.

ब) अंतिम गुणवत्ता यादी ही वस्तुनिष्ठ परीक्षेतील व मुलाखतीतील एकत्रित गुणांवर आधारीत राहील.

-: अभ्यासक्रम :-

विषयांशी संबंधित घटक यामध्ये खालील घटक व उपघटकांचा समावेश असेल.

अ.क्र.	घटक व उपघटक
1	**Engineering Mathematics** ● Discrete Mathematics: Propositional and first-order logic, sets, relations, functions, partial orders, lattices, monoids, groups, graph theory (connectivity, matching, coloring), and combinatorics (counting, recurrence relations, generating functions). ● Linear Algebra: Matrices, determinants, systems of linear equations, eigenvalues, eigenvectors, and LU decomposition. ● Calculus: Limits, continuity, differentiability, maxima and minima, mean value theorem, integration. ● Probability and Statistics: Probability distributions (Uniform, Normal, Exponential, Poisson, Binomial), mean, median, mode, standard deviation, Bayes theorem.
2	**Digital Logic** ● Boolean algebra. ● Combinational and sequential circuits. ● Minimization techniques: Karnaugh maps, Quine-McCluskey. ● Number representations and computer arithmetic (fixed-point and floating-point). ● Specific combinational circuits: adders, multiplexers, decoders. ● Sequential circuits: flip-flops, counters, registers.
3	**Computer Organization and Architecture** ● Machine instructions and addressing modes. ● ALU, data-path, and control unit. ● Instruction pipelining and pipeline hazards. ● Memory hierarchy: cache, main memory, secondary storage. ● I/O interface: interrupt and DMA modes.
4	**Programming and Data Structures** ● Programming in C. ● Recursion. ● Arrays, stacks, queues, linked lists, trees, binary search trees, binary heaps, graphs.

5	**Algorithms** ● Searching, sorting, hashing. ● Asymptotic worst-case time and space complexity. ● Algorithm design techniques: greedy, dynamic programming, divide-and-conquer. ● Graph algorithms: traversals, minimum spanning trees, shortest paths.
6	**Theory of Computation** ● Regular expressions and finite automata. ● Context-free grammars and push-down automata. ● Regular and context-free languages, pumping lemma. ● Turing machines and undecidability.
7	**Compiler Design** ● Lexical analysis, parsing, syntax-directed translation. ● Runtime environments. ● Intermediate code generation. ● Local optimization. ● Data flow analyses: constant propagation, liveness analysis, common subexpression elimination.
8	**Operating System** ● System calls, processes, threads. ● Inter-process communication, concurrency, and synchronization. ● Deadlock. ● CPU and I/O scheduling. ● Memory management and virtual memory. ● File systems.
9	**Databases** ● ER-model. ● Relational model: relational algebra, tuple calculus, SQL. ● Integrity constraints, normal forms. ● File organization, indexing (e.g., B and B+ trees). ● Transactions and concurrency control.
10	**Computer Networks** ● Concept of layering: OSI and TCP/IP Protocol Stacks. ● Basics of packet, circuit, and virtual circuit switching. ● Data link layer: framing, error detection, Medium Access Control, Ethernet bridging. ● Routing protocols: shortest path, flooding, distance vector, and link state routing. ● Fragmentation and IP addressing: IPv4, CIDR notation. ● Basics of IP support protocols: ARP, DHCP, ICMP. ● Network Address Translation (NAT). ● Transport layer: flow control and congestion control, UDP, TCP, sockets. ● Application layer protocols: DNS, SMTP, HTTP, FTP, Email.

दिनांक – २१ जुलै, २०२५

अवर सचिव
महाराष्ट्र लोकसेवा आयोग